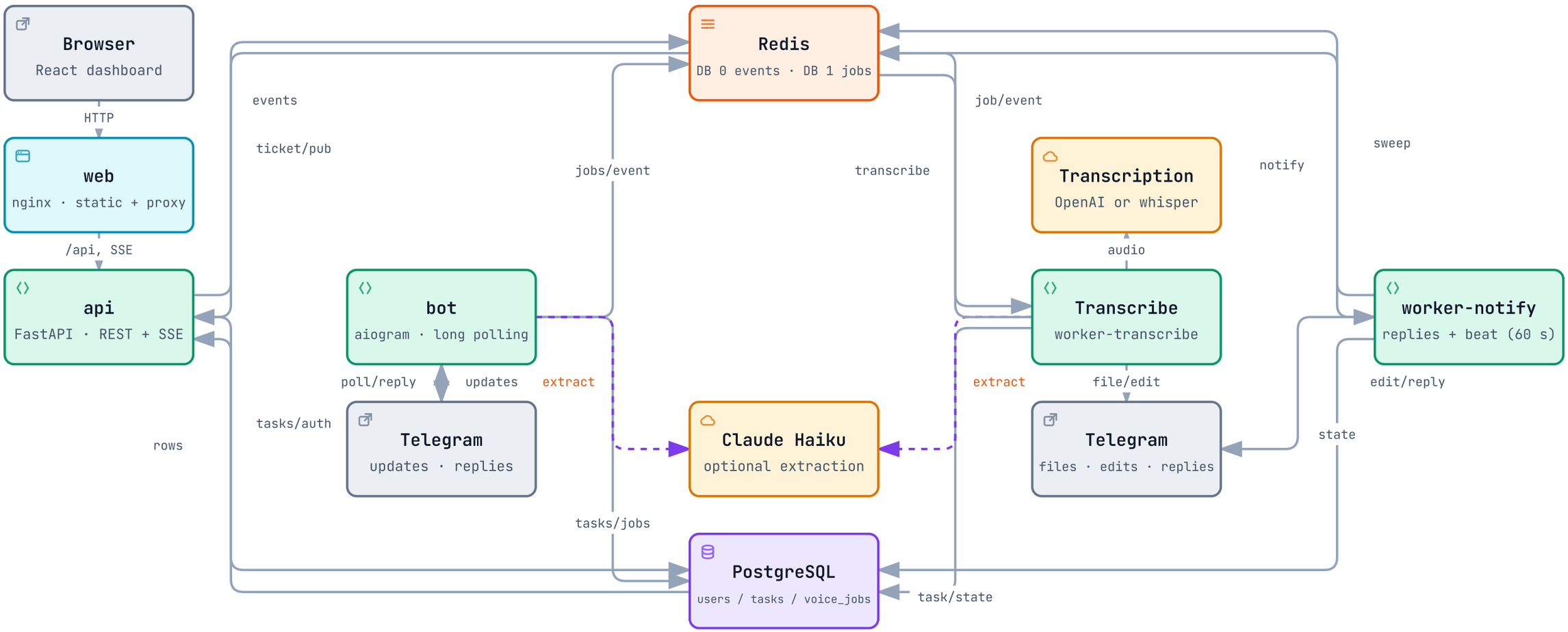
AI Task Bot — System overview



Legend

Frontend · Backend · Database · Cloud · Message bus · External

One backend image, six services

• migrate, api, bot, worker-transcribe, worker-notify, seed. Migrate: alembic upgrade head, exit 0 before apps. Only web publishes 127.0.0.1:WEB_PORT (8080).

One service, two boxes or roles

• Telegram repeats one Bot API. Redis DB 1: transcription, notification. DB 0: events:user:{id}, tickets, bot lock, heartbeat. PostgreSQL: users, inbound_messages, tasks, voice_jobs.

What is optional

• OpenAI is default; no key means voice unavailable. TRANSCRIPTION_PROVIDER selects OpenAI or CPU whisper.cpp (COMPOSE_PROFILES=whisper). Claude: TASK_EXTRACTOR=claude + key, in bot/worker/seed. seed: tools profile.